

HW4

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MATH 339 — Gonzaga University

compiled with L^AT_EX

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Problem 11

$$c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + c_3 \mathbf{v}_3 = \mathbf{v}$$

$$\begin{aligned} & \left[\begin{array}{ccc|c} 2 & 1 & -1 & 3 \\ -3 & 6 & -1 & -17 \\ 4 & -1 & 2 & 17 \\ 1 & 2 & 3 & 7 \end{array} \right] \xrightarrow{R_1 \leftrightarrow R_4} \left[\begin{array}{ccc|c} 1 & 2 & 3 & 7 \\ -3 & 6 & -1 & -17 \\ 4 & -1 & 2 & 17 \\ 2 & 1 & -1 & 3 \end{array} \right] \\ & \xrightarrow[\begin{smallmatrix} R_3 - 4R_1 \\ R_4 - 2R_1 \end{smallmatrix}]{R_2 + 3R_1} \left[\begin{array}{ccc|c} 1 & 2 & 3 & 7 \\ 0 & 12 & 8 & 4 \\ 0 & -9 & -10 & -11 \\ 0 & -3 & -7 & -11 \end{array} \right] \xrightarrow{R_2/4} \left[\begin{array}{ccc|c} 1 & 2 & 3 & 7 \\ 0 & 3 & 2 & 1 \\ 0 & -9 & -10 & -11 \\ 0 & -3 & -7 & -11 \end{array} \right] \\ & \xrightarrow[\begin{smallmatrix} R_4 + R_2 \end{smallmatrix}]{\begin{smallmatrix} R_3 + 3R_2 \end{smallmatrix}} \left[\begin{array}{ccc|c} 1 & 2 & 3 & 7 \\ 0 & 3 & 2 & 1 \\ 0 & 0 & -4 & -8 \\ 0 & 0 & -5 & -10 \end{array} \right] \xrightarrow{R_3/(-4)} \left[\begin{array}{ccc|c} 1 & 2 & 3 & 7 \\ 0 & 3 & 2 & 1 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & -5 & -10 \end{array} \right] \\ & \xrightarrow{R_4 + 5R_3} \left[\begin{array}{ccc|c} 1 & 2 & 3 & 7 \\ 0 & 3 & 2 & 1 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{array} \right] \end{aligned}$$

consistent ($0 \ 0 \ 0 = 0$), so $\mathbf{v}$ is a linear combination.

$$c_3 = 2$$

$$3c_2 + 2(2) = 1 \Rightarrow c_2 = -1$$

$$c_1 + 2(-1) + 3(2) = 7 \Rightarrow c_1 = 3$$

$$\boxed{\mathbf{v} = 3\mathbf{v}_1 - \mathbf{v}_2 + 2\mathbf{v}_3}$$

Problem 13

$$c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + c_3 \mathbf{v}_3 = \mathbf{v}$$

$$\left[\begin{array}{ccc|c} 3 & 0 & -1 & 3 \\ 1 & -1 & 2 & 0 \end{array} \right] \xrightarrow{R_1 \leftrightarrow R_2} \left[\begin{array}{ccc|c} 1 & -1 & 2 & 0 \\ 3 & 0 & -1 & 3 \end{array} \right] \xrightarrow{R_2 - 3R_1} \left[\begin{array}{ccc|c} 1 & -1 & 2 & 0 \\ 0 & 3 & -7 & 3 \end{array} \right]$$

$$\xrightarrow{R_2/3} \left[\begin{array}{ccc|c} 1 & -1 & 2 & 0 \\ 0 & 1 & -7/3 & 1 \end{array} \right] \xrightarrow{R_1+R_2} \left[\begin{array}{ccc|c} 1 & 0 & -1/3 & 1 \\ 0 & 1 & -7/3 & 1 \end{array} \right]$$

$c_3 = t$ free

$$c_1 = 1 + \frac{1}{3}t$$

$$c_2 = 1 + \frac{7}{3}t$$

$$c_3 = t$$

$$\text{solution} = \left\{ (1, 1, 0) + t \left(\frac{1}{3}, \frac{7}{3}, 1 \right) : t \in \mathbb{R} \right\}$$